

VisionClaw Agent Platform

Comprehensive Features & Capabilities

AI Buddy LLC — Illinois

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Platform Overview

VisionClaw is an agentic AI platform designed as a fully autonomous AI corporation. It features 14 specialized AI personas operating as a complete corporate team, governed by 40 rules based on NIST, OWASP, and Singapore IMDA standards, implementing the Claude Opus 4.6 Agentic Spec. Agents handle tasks autonomously and escalate only mission-critical issues to human owners. Built for multi-tenancy, cost-effectiveness, and robust security.

Platform Stats (April 2026)

- 146 server-side TypeScript files
- 38 frontend pages + 110 client TypeScript/TSX files
- 67 database tables (PostgreSQL + pgvector)
- 97 built-in AI tools + custom tool creation
- 23 active skills across business and platform operations
- 14 AI personas (complete corporate team)
- 36+ models across 8+ providers
- 40 governance rules across 7 categories
- 11 autonomous research programs (5 nightly + 6 weekly business)
- 7 research schedules
- 13 heartbeat tasks (5 active, 8 approved/available)
- Multi-tenant architecture with full data isolation

14 AI Personas — Corporate Team

VisionClaw (CEO) — System orchestrator and platform coordinator

Felix (COO) — Chief Operating Officer — user-facing assistant, task orchestration, multi-step delegation

Forge (CTO) — Chief Technology Officer — code generation, architecture, technical analysis

Teagan (CMO) — Chief Marketing Officer — content creation, social media, brand strategy

Blueprint (VP Engineering) — Engineering planning, system design, technical documentation

Chief of Staff — Administrative operations, scheduling, cross-team coordination

Scribe (Content Director) — Long-form content, documentation, copywriting, communications

Proof (QA Director) — Quality assurance, adversarial review, output verification

Radar (Intelligence) — Market research, competitive intelligence, trend analysis

Neptune (Wellness) — Health & wellness guidance, emotional eating interventions

Apollo (Strategy) — Business strategy, revenue optimization, pricing analysis

Atlas (Finance) — Financial analysis, reporting, budget management

Cassandra (Risk) — Risk assessment, financial modeling, predictive analytics

Luna (Legal/Compliance) — Legal research, compliance frameworks, regulatory analysis

97 Built-in AI Tools

Communication & Email

- send_email, read_inbox, whatsapp_send, discord_send, telegram_send

Research & Intelligence

- web_search (multi-provider), scrape_url (Firecrawl), trend_research (Reddit/HN/X/Polymarket), deep_research, strategic_interview (Socratic requirement gathering)

Document & Content Creation

- generate_document (PDF/DOCX/TXT), create_slides (PowerPoint via 2slides), render_diagram (Mermaid to PNG), generate_social_image (AI images), produce_video (TTS narration + MP4), generate_audio (OpenAI/ ElevenLabs TTS), generate_chart (bar/line/pie/area), generate_dashboard (live HTML dashboards)

Code & Development

- execute_code (sandboxed), code_review, create_code_proposal, manage_project

Agentic Operations

- delegate_task (5-level depth), orchestrate (DAG decomposition), manage_heartbeat, manage_governance, manage_trust, manage_autonomy, manage_watchlist, export_persona, sync_personas

Virtual Browser

- browser_navigate, browser_screenshot, browser_fill_form, browser_click, browser_extract

Google Workspace

- upload_to_drive, list_drive_files, create_drive_folder

Finance & Markets

- stock_search, stock_data (OHLCV), market_news, market_overview

System & Configuration

- manage_skills, manage_models, manage_channels, manage_mcp, backup_system, vibevoice_transcribe (speech-to-text with diarization)

Autonomous Operations

Heartbeat Engine (Autonomous Nervous System)

Runs scheduled background tasks without human intervention:

- Self-Reflection — Agents review performance and adjust behavior
- Memory Consolidation — Compacts and optimizes agent memory
- Dream Memory Consolidation — Background 'sleep' engine (merges, archives, promotes, summarizes every

6h)

- Model Scout — Discovers and evaluates new AI models
- Daily Cloud Backup — Automatic database + config backup to Google Drive
- Decision Analysis, Trend Forecasting, Process Optimization, Process Governor — Available strategic engines
- Quarterly Governance Research & Model Registry Refresh — Available regulatory/model scanners
- 100% uptime success rate (93/93 heartbeat runs in last 7 days)

Nightly Autoresearch System (Karpathy-inspired)

11 autonomous research programs run on schedule. Each spawns 5-15 experiments per session using a keep/discard loop:

- Emotional Eating Crisis Interventions
- AI Buddy Content Marketing Pipeline
- Competitive Intelligence — Weight Loss & Emotional Eating Market
- AI Buddy Revenue & Pricing Strategy
- Daily Companion Message Library
- AI Buddy Legal & Compliance Framework
- Nightly AI Model & Provider Intelligence
- Nightly AI Tools & Techniques Scanner
- Nightly Competitive Platform Analysis
- Nightly Agent Architecture Research
- Nightly Security & Safety Intelligence

Last 4 days: 32 sessions, 352 experiments, 92 findings kept (26%), 114 knowledge entries auto-injected.

Auto-Deposit Intelligence Loop

Research findings auto-route to correct project folder AND knowledge base with vector embeddings. Business research goes to AI Buddy project, platform research goes to VisionClaw project. The system gets smarter with every session.

Adaptive Research Backpressure

Self-healing throttle: Level 1 (2x slower), Level 2 (3x + reschedule), Level 3 (pause 5min). Max 2 concurrent sessions. Never overwhelms the database.

Claude Code-Inspired Features (New — March/April 2026)

Six major features inspired by Claude Code's agentic architecture, all completed:

1. DreamTask Memory Consolidation Engine

Background 'sleep' engine running every 6 hours when idle. Reviews session memories, merges duplicates, archives stale entries, promotes important findings, creates cross-topic summaries. Keeps the knowledge base clean and efficient without human intervention.

2. Adversarial Verification Agent (Auto-QA / De-Sloppify)

After every delegation, Proof (persona 8) automatically reviews output quality. Scores completeness, accuracy, clarity, professionalism (1-10). Color-coded verdict badges in live feed. Non-blocking — never delays delegation returns.

3. Agent Live Status Summaries

Real-time SSE delegation event feed with animated bubbles, agent icons, depth indicators, timestamps, and per-task cost tracking. Voice narration speaks activity in plain English via Chrome speech synthesis (free). Fully tenant-isolated.

4. Skillify — Session-to-Skill Extraction

Extracts reusable execution patterns from successful multi-tool tasks. Stored as 'instincts' with confidence scores. After 3+ observations above 70%, instincts graduate to permanent knowledge in system prompts. Agents improve through experience.

5. Enhanced Stuck Detection and Diagnostics

Multi-signal analysis: long-running ops, repetitive loops, consecutive failures. Auto-diagnostic with remediation suggestions. Tenant-safe metadata stripping prevents cross-tenant leakage.

6. LLM-Judged Memory Relevance Scoring

GPT-4.1 Mini selects best 5-7 knowledge entries via fast side-query (2s timeout, 30s cache). Smart filtering skips generic docs for active tools, preserves warnings. Vector similarity fallback. Agents always get the most relevant knowledge.

Security & Governance

40-Rule Governance Engine

7 categories, 25 condition evaluators, emergency Kill Switch, NIST/OWASP/IMDA governance framework knowledge base.

Trust Score System

9 categories per persona: purpose_adherence, tool_compliance, delegation_efficiency, external_comms, content_quality, research_accuracy, financial_accuracy, legal_soundness, code_reliability. Autonomy levels: full_auto (80+), notify_after (60+), approve_before (<60).

Agency Expansion Framework (6 Tiers)

- Evaluators (9 types) — Continuous performance assessment
- Trust Score Engine (9 categories) — Earned autonomy progression
- Proactive Initiative Engine — PAB budgets for autonomous actions
- Express Lanes (12 approved) — Pre-approved action patterns
- Environmental Awareness (8 scan types) — Contextual intelligence
- Collective Intelligence (4 protocols) — Multi-agent coordination

Security Measures

- Multi-tenant data isolation at every layer
- Admin PIN authentication (HMAC-SHA256)
- Timing-safe cryptographic comparisons
- Helmet CSP headers
- Provider Key Proxy (no direct key exposure)
- IronClaw-inspired SafetyLayer
- Per-tool rate limiting (sliding window per tenant)
- Circuit breaker on tool execution
- Hallucination detection in outputs

Recent Security Fixes (8 Critical)

- Admin guard on backup/import endpoints
- LiveCanvas XSS prevention (iframe sandbox)
- Knowledge PATCH IDOR fix
- Heartbeat approve/reject requires admin
- Capabilities cache made per-tenant
- Auto-QA result field bug fixed
- Auth token key standardized
- Stuck diagnostics metadata stripped

Smart AI Model Routing

36+ Models Across 8+ Providers

OpenAI (GPT-4.1, GPT-4.1 Mini, o4-mini), Anthropic (Claude Opus 4.6, Sonnet 4, Sonnet 4.6), Google (Gemini 3 Flash, 2.5 Flash, 3.1 Pro), xAI (Grok), Perplexity (Sonar), OpenRouter (DeepSeek, MiniMax, Qwen, Llama, Kimi, Nemotron, Mistral), Claude Runner bridge.

OAuth-First Cost Optimization

Subscription-first routing: OAuth tokens from ChatGPT Plus and Google Gemini used as primary inference (\$0 marginal). Claude Runner for \$0 Anthropic. Auto-routing: free != cheap != paid.

Task Complexity Tiers

- Fast: Gemini 3 Flash != Gemini 2.5 Flash != GPT-4.1 Mini
- Balanced: Gemini 3 Flash != GPT-4.1 != Sonnet 4
- Powerful: Gemini 3.1 Pro != GPT-4.1 != Opus 4.6
- Reasoning: Gemini 3.1 Pro != o4-mini != Opus 4.6

Memory & Intelligence System

- Three-Tier Semantic Memory (Episodic, Semantic, Procedural)
- Vector Knowledge Library (pgvector, text-embedding-3-small)
- BM25/Vector/Hybrid Document Search
- Zero-Loss Compaction for long conversations
- DreamTask Consolidation (6h cycle)
- Instinct Learning (pattern extraction != graduation)
- LLM-Judged Relevance Scoring (GPT-4.1 Mini)
- Cross-persona knowledge retrieval

Frontend & UX

- 38 pages: Dashboard, Chat, Agents, Settings, Reports, Projects, Research, Governance, etc.
- Dark mode with full theme support
- Real-time delegation activity feed with voice narration
- Interactive charts and live dashboards in chat
- 3-step onboarding, keyboard shortcuts, error boundary
- Grouped sidebar with auto-named conversations
- Stop generating, regenerate, scroll-to-bottom

External Integrations

- OpenAI — GPT models + TTS + embeddings
- Anthropic — Claude models + Runner bridge
- Google Gemini — OAuth + integration
- Google Drive — File storage, sharing, backup
- Stripe — Payments, subscriptions, BYOK billing
- Coinbase — Crypto payments via Commerce API
- ElevenLabs — TTS (Creator, 110K chars/month, 23 voices)
- Firecrawl — Web scraping, crawling, site mapping
- AgentMail — Tenant email system

- WhatsApp, Discord, Telegram — Messaging bots
- 2slides — PowerPoint generation
- Replit Object Storage — Tenant file storage

Technology Stack

Frontend

React 18, Vite, TypeScript, shadcn/ui, TailwindCSS, Wouter, TanStack Query v5, Framer Motion, Recharts

Backend

Express.js, TypeScript, Drizzle ORM, Zod, Helmet, SSE streaming

Database

PostgreSQL with pgvector, 67 tables, Drizzle ORM

AI/ML

36+ models, text-embedding-3-small, pgvector semantic search, GPT-4.1 Mini relevance scoring